

# A two-valent Miniowitz map onto the punctured plane

## Explicit answer to the omitted-set construction problem

Autonomous solve-first investigation, lane 12

13 August 2026

### Abstract

Adamowicz and González ask for quasiregular maps satisfying their Möbius-invariant multiplicity condition (M) and Miniowitz estimate (Min), but not the radial omitted-set hypothesis, preferably maps omitting only the origin. The elementary holomorphic map

$$F(z) = \left( \frac{1+z}{1-z} \right)^3$$

does exactly this. It is 1-quasiregular, maps the disk onto  $\mathbb{C} \setminus \{0\}$ , has valence at most two, satisfies (M) with exponent  $a = 0$ , and satisfies (Min) after every conformal recentering with  $C = 1$  and  $\alpha = 3$ . More generally every integer power at least three works. We also record the exact Hardy threshold:  $F \in \mathcal{H}^p$  precisely for  $0 < p < 1/3$ .

## 1 The source question

The source is T. Adamowicz and M. J. González, *Hardy spaces and quasiregular mappings* [1]. Its arXiv identifier is 2309.12947. For a quasiregular map  $f : B^n \rightarrow \mathbb{R}^n$ , condition (M) requires constants  $A > 0$  and  $0 \leq a < n - 1$  such that

$$\sup_{x \in B^n} N(f \circ T_x^{-1}, B(0, r)) \leq \frac{A}{(1-r)^a} \quad (0 < r < 1), \quad (1)$$

where  $T_x$  is a conformal automorphism taking  $x$  to zero. Estimate (Min) requires, uniformly after every such recentering  $g = f \circ T_x^{-1}$ ,

$$\frac{1}{C} \left( \frac{1-r}{1+r} \right)^\alpha \leq \frac{|g(y)|}{|g(0)|} \leq C \left( \frac{1+r}{1-r} \right)^\alpha, \quad r = |y|. \quad (2)$$

The sufficient hypothesis preceding the question assumes that  $f$  omits a set meeting every sphere centered at the origin. The authors ask for maps satisfying (1) and (2) for which that hypothesis is unnecessary, “omit[ting] e.g. only the origin.” The construction below meets that strongest requested range condition.

## 2 The construction

**Theorem 1** (Cayley-power family). *For every integer  $m \geq 3$ , define*

$$H(z) = \frac{1+z}{1-z}, \quad F_m(z) = H(z)^m \quad (z \in \mathbb{D}). \quad (3)$$

*Then:*

1.  $F_m$  is 1-quasiregular and  $F_m(\mathbb{D}) = \mathbb{C} \setminus \{0\}$ ;
  2. its valence is at most  $\lceil m/2 \rceil$ , so it satisfies (M) with  $a = 0$  and  $A = \lceil m/2 \rceil$ ;
  3. after every conformal disk recentering, (Min) holds with  $C = 1$  and  $\alpha = m$ .
- In particular,  $F_3$  is an at-most-two-valent explicit solution of the source problem.*

*Proof.* The Cayley map  $H$  is a conformal bijection from  $\mathbb{D}$  onto the right half-plane  $\mathbb{H} = \{u : \operatorname{Re} u > 0\}$ . The argument interval of  $u^m$  as  $u$  ranges over  $\mathbb{H}$  is  $(-m\pi/2, m\pi/2)$ . Its length is  $m\pi > 2\pi$ , so it contains a representative of every argument modulo  $2\pi$ . Every positive modulus is also attained. Hence  $u \mapsto u^m$  maps  $\mathbb{H}$  onto  $\mathbb{C} \setminus \{0\}$ , proving the exact range assertion. Holomorphicity makes  $F_m$  1-quasiregular.

For fixed  $w \neq 0$ , the arguments of its  $m$ th roots differ by  $2\pi/m$ . The open right-half-plane argument interval has length  $\pi$ , and therefore contains at most  $\lceil m/2 \rceil$  of these roots. Since  $H$  is injective, the same is a global valence bound for  $F_m$ . Precomposition by  $T_x^{-1}$  does not change it; every restriction to  $B(0, r)$  has no larger multiplicity. Thus (1) holds with  $a = 0$ .

It remains to prove the uniform estimate, for which no distortion theorem is needed. Fix a disk automorphism  $T$ , put  $g = F_m \circ T^{-1}$ ,  $z = T^{-1}(y)$ ,  $z_0 = T^{-1}(0)$ ,  $u = H(z)$ , and  $v = H(z_0)$ . Conformal invariance of the pseudohyperbolic metric gives

$$r = |y| = \left| \frac{u - v}{u + \bar{v}} \right|. \quad (4)$$

Consequently

$$||u| - |v|| \leq |u - v| = r|u + \bar{v}| \leq r(|u| + |v|).$$

Rearranging this inequality in both directions yields

$$\frac{1 - r}{1 + r} \leq \frac{|u|}{|v|} \leq \frac{1 + r}{1 - r}. \quad (5)$$

Since  $|g(y)|/|g(0)| = (|u|/|v|)^m$ , raising (5) to the  $m$ th power is exactly (2) with  $C = 1$  and  $\alpha = m$ .  $\square$

Because the range of  $F_m$  is all of  $\mathbb{C} \setminus \{0\}$ , every set omitted by  $F_m$  is contained in  $\{0\}$ . It therefore cannot meet any centered circle of positive radius. Thus the construction does not merely use a smaller omitted set; it fails the source's radial omitted-set condition maximally.

### 3 Exact Hardy membership

**Corollary 2.** *For  $m \geq 3$  and  $p > 0$ ,*

$$F_m \in \mathcal{H}^p(\mathbb{D}) \iff mp < 1.$$

*In particular,  $F_3 \in \mathcal{H}^p$  exactly when  $0 < p < 1/3$ .*

*Proof.* The only boundary singularity is at  $z = 1$ . Uniformly for  $r$  close to one and  $|t|$  small,

$$|F_m(re^{it})|^p = \frac{|1 + re^{it}|^{mp}}{|1 - re^{it}|^{mp}} \asymp ((1 - r)^2 + t^2)^{-mp/2}.$$

The integrals of the right-hand side over a fixed small interval are uniformly bounded as  $r \uparrow 1$  exactly when  $mp < 1$ . At  $mp = 1$  they grow logarithmically, and for  $mp > 1$  they grow as a negative power of  $1 - r$ . Away from  $t = 0$  the integrand is uniformly bounded.  $\square$

### 4 Verification and literature scope

The proof checks the source's stronger Möbius-invariant form of (M), not only multiplicity in centered Euclidean balls. Equation (4) also establishes (Min) separately for every recentering, with constants independent of the center. No boundary-value theorem is used.

A bounded search on 13 August 2026 used the exact source sentence, the paper title, Miniowitz-class terminology, bounded multiplicity, and the displayed Cayley power. It found the source and later work noting abstractly that the Miniowitz class includes bounded-multiplicity maps, but no explicit Cayley-power construction. The source itself cites the general bounded-multiplicity estimate. Accordingly, this packet claims an explicit full answer and elementary sharp verification, not priority for that general structural fact.

### References

- [1] T. Adamowicz and M. J. González, *Hardy spaces and quasiregular mappings*, arXiv:2309.12947.